

ECO-3554/ECO-4554 Algorithmic Economics

Monsoon Semester 2026 (Aug–Dec)

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Lectures:	Tu-Th 10:10 AM –11:40 AM in AC-04-LR-303
Course Webpage:	goelsumit.com/algoecon.html

Course Description

The course is an introduction to algorithmic questions in economic design. We will cover resource allocation, fair division, social choice, and auctions. In these contexts, we will formulate design goals, discuss rules that achieve them, and study workarounds for when the goals are incompatible. Social goals of efficiency and fairness, and private goals like revenue maximization, will be considered under incentive, computational, and communicational constraints.

Learning Objectives

- Become familiar with the tools of microeconomics that guide modern economic design.
- Identify incentive issues, and develop a working knowledge of mechanism design.
- Recognize the computational and communicational issues in economic environments.

Prerequisites

There are no formal course prerequisites. The content of the course is, however, formal and mathematical. Students should be comfortable with formal mathematical proofs and will be expected to write proofs on their own.

Evaluation

This course is lecture based, and student assessment focuses on problem solving. The final course grade will be composed of the following components:

Component	Weight
Assignments/Quizzes	40%
Midterm exam	30%
Final exam	30%

The numeric score will be converted to a letter grade according to the following scheme:

A	A−	B+	B	B−	C+	C	C−	D+	D	F
>95	90–95	85–90	80–85	75–80	70–75	65–70	60–65	55–60	50–55	<50

Attendance

Regular attendance in this course is expected and highly recommended.

Course Outline

Weeks	Topics
Week 1	Aggregating preferences: efficiency, incentives, impossibility
Weeks 2–3	Escaping impossibilities: domain restrictions, complexity barriers
Weeks 4–5	Quasilinear domain: VCG, approximation ratio
Weeks 6–7	Auctions: revenue maximization, combinatorial auctions
Weeks 8–9	Fair division (divisible goods): cake cutting, rent division
Weeks 10–11	Fair division (indivisible goods): Nash welfare, randomization
Weeks 12–13	Cooperative games: Shapley value, feature attribution in ML

Readings

You may find the following sources useful, which I use to prepare the lectures:

- [Lecture notes](#) by Fedor Sandomirskiy
- [Lecture notes](#) by Ariel Procaccia

I also borrow ideas from Federico Echenique's [course](#) on Algorithmic Economics.

This syllabus is subject to change at the discretion of the instructor.